TEL AVIV UNIVERSITY | FACULTY OF ENGINEERING

Linear Algebra

Week 1

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1 Sets and the language of mathematics

1.1 Describing a set

A *set* is a collection of objects. Its members may be numbers, geometric objects, or even other sets. For example,

$$A = \{2, 3, 8\}$$

is a set with three elements. We can describe a set by listing its elements, by explaining in words which objects belong to it, or by giving a condition that determines membership.

Example 1.1 (Several ways to specify a set)**.** 1. The list $\{2, 3, 8\}$ specifies a set completely. It contains 3 but does not contain 5.

2. The notation $\{1, 2, \ldots, 12\}$ describes the first twelve positive integers. It contains 9 but not 14.
3. An abbreviated list can be ambiguous. Does $\{3, 9, \ldots\}$ mean the positive powers of 3, namely $\{3, 9, 27, 81, \ldots\}$, or the positive odd multiples of 3, namely $\{3, 9, 15, 21, \ldots\}$? A verbal description removes the ambiguity.
4. A set need not contain numbers: $\{\text{red}, \text{green}, \text{blue}\}$ is a set of colors.
5. The *empty set*, denoted $\varnothing$, has no elements.
6. The set $C = \{\{2, 3, 8\}, \{2, 5\}\}$ has two elements, and both are sets. The set $\{3, 8\}$ is not one of those two elements.
7. When our universe is the real numbers, $\{x \mid x < 0\}$ is the set of negative real numbers. The vertical bar means "such that." To make the universe explicit, write $\{x \in \mathbb{R} \mid x < 0\}$.

The order of the elements does not matter, and repetitions do not create new elements. Thus

$$\{2, 3, 8\} = \{8, 2, 3\} = \{2, 3, 2, 8, 8\}.$$

Definition 1.1 (Membership)**.** Write $a \in A$ when a is an element of the set A. Write $a \notin A$ when it is not.

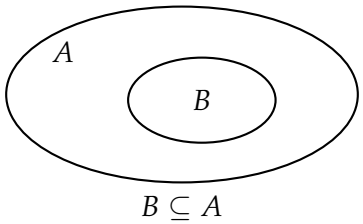
For example, $8 \in \{2, 3, 8\}$, whereas $6 \notin \{2, 3, 8\}$.

1.2 Subsets and the difference between $\in$ and $\subseteq$

Definition 1.2 (Inclusion)**.** For sets A and B, we say that B is a *subset* of A, and write $B \subseteq A$, if every element of B belongs to A. In symbols,

$$B \subseteq A \quad \Longleftrightarrow \quad \text{for every } b \in B,\ b \in A.$$

The notation $B \subset A$ also permits equality. We mainly use $\subseteq$ to make that convention visible. If the sets are unequal, $B \subsetneq A$ denotes a *proper* subset. Other books sometimes use $\subset$ for proper inclusion, so check the convention before reading it.



$B \subseteq A$

Example 1.2 (Membership is not inclusion)**.** 1. $\{2,8\} \subseteq \{2,3,8\}$, and the inclusion is proper. But $\{2,5\} \nsubseteq \{2,3,8\}$, because 5 is missing from the second set.

2. Every set is a subset of itself: $A \subseteq A$.
3. If $A = \{2,3,8\}$, then $B = \{a \in A \mid a \text{ is even}\} = \{2,8\}$.
4. For $\mathcal{F} = \{\{2,3,8\},\{2,8\}\}$, both A and B are elements of $\mathcal{F}$. However, $\{2,3\} \notin \mathcal{F}$: it is not one of the two listed elements.
5. Let $D = \{2,5\}$ and $E = \{\{2,5\},\{3,7\}\}$. Then $D \in E$, but $D \nsubseteq E$, because neither the number 2 nor the number 5 is an element of E.

To check $D \in E$, look for the single object D among the elements of E. To check $D \subseteq E$, examine every element of D separately.

Remark 1.3. The empty set is a subset of every set: $\varnothing \subseteq A$. There is no element of $\varnothing$ that could violate the inclusion condition. On the other hand, $\varnothing \in A$ is a different statement and depends on A. For example, $\varnothing \in \{\varnothing, 2\}$, but $\varnothing \notin \{2,3\}$.

1.3 The power set

Definition 1.4 (Power set)**.** The *power set* of a set A, denoted $P(A)$, is the set of all subsets of A:

$$P(A) = \{B \mid B \subseteq A\}.$$

Example 1.3. For $A = \{2,3,8\}$,

$$P(A) = \{\varnothing, \{2\}, \{3\}, \{8\}, \{2,3\}, \{2,8\}, \{3,8\}, \{2,3,8\}\}.$$

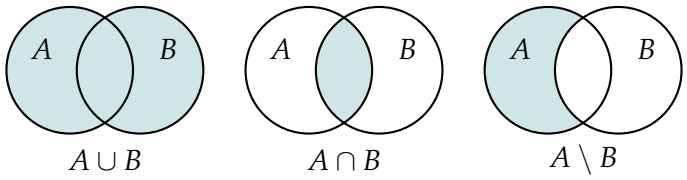
This power set has eight elements. Notice that its elements are subsets, including both $\varnothing$ and A itself.

1.4 Union, intersection, difference, and complement

Definition 1.5 (Operations on sets)**.** For sets A and B, define

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\},$$
$$A \cap B = \{x \mid x \in A \text{ and } x \in B\},$$
$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}.$$

These are the *union*, *intersection*, and *difference*, respectively. The word "or" here is inclusive: an element in both sets belongs to the union too.



Example 1.4. Let $A = \{2,3,8\}$ and $B = \{3,5\}$. Then

$$A \cup B = \{2,3,5,8\}, \qquad A \cap B = \{3\},$$
$$A \setminus B = \{2,8\}, \qquad B \setminus A = \{5\}.$$

A set with exactly one element, such as $\{3\}$, is called a *singleton*.

Now let $C = \{\{2,3\},\{2,7\}\}$ and $D = \{\{3,5\},\{7\}\}$. There is no identical set listed in both collections, so $C \cap D = \varnothing$. Such sets are called *disjoint*. We do not intersect the elements with one another: $\{3\}$ is not an element of either collection.

Definition 1.6 (Complement relative to a universe)**.** Suppose all sets under consideration are subsets of a fixed set U, called the *universe*. For $B \subseteq U$, its complement is

$$B^c = U \setminus B.$$

The universe matters. For example, if $U = \{0, 1, 2, \ldots\}$, then $\mathbb{N}^c = \{0\}$, using $\mathbb{N} = \{1, 2, \ldots\}$. If instead the universe is $\mathbb{Z}$, the complement of $\mathbb{N}$ is $\{\ldots, -2, -1, 0\}$. When the universe is unclear, write the set difference explicitly.

Example 1.5 (Course enrollment)**.** Let U be the students in a program, A the students taking algebra, and B those taking physics. Then $A \cap B$ describes students taking both courses, $A \setminus B$ describes those taking algebra without physics, and $(A \cup B)^c$ describes those taking neither course. A student taking both courses still appears only once in the set $A \cup B$.

1.5 The standard number systems

We use $\mathbb{N} = \{1, 2, 3, \ldots\}$, so $0 \notin \mathbb{N}$.

Symbol	Name	Description
$\mathbb{N}$	Natural numbers	$\{1, 2, 3, \ldots\}$, used for counting.
$\mathbb{Z}$	Integers	$\{\ldots, -2, -1, 0, 1, 2, \ldots\}$.
$\mathbb{Q}$	Rational numbers	Fractions m/n with $m, n \in \mathbb{Z}$ and $n \neq 0$.
$\mathbb{R}$	Real numbers	The numbers on the real line. This description is intuitive rather than a construction of $\mathbb{R}$.
$\mathbb{C}$	Complex numbers	Numbers $a + bi$ with $a, b \in \mathbb{R}$ and $i^2 = -1$.

Negative integers allow us to describe debts or solve equations such as $x + 7 = 3$. Fractions allow us to solve equations such as $3x = 2$. In a rational number, a negative denominator can be moved to the numerator: for example,

$$\frac{6}{-15} = -\frac{2}{5}.$$

Every rational number has a unique *canonical representation* m/n with $m \in \mathbb{Z}$, $n \in \mathbb{N}$, and m, n relatively prime. "Relatively prime" means that their greatest common divisor is 1; for $m = 0$, the canonical representation is $0/1$.

The real numbers are not all rational. Examples of irrational real numbers include $\sqrt{2}$, $\sqrt{3}$, π, and e. We will prove the irrationality of $\sqrt{3}$ below. A real number that is not rational is called *irrational*.

1.6 Proving equality by two inclusions

To prove that two sets are equal, it is usually useful to show both inclusions:

$$A \subseteq B \quad \text{and} \quad B \subseteq A.$$

The first inclusion rules out elements found only in A; the second rules out elements found only in B.

Example 1.6 (Rational numbers as solutions of linear equations)**.** Let $A = \mathbb{Q}$, and define

$$B = \{x \in \mathbb{R} \mid \text{there exist } a, b \in \mathbb{Z},\ (a, b) \neq (0, 0),\ ax + b = 0\}.$$

We claim that $A = B$.

First prove $A \subseteq B$. If $x \in A$, write $x = m/n$ with $m, n \in \mathbb{Z}$ and $n \neq 0$. Then $nx - m = 0$. Taking $a = n$ and $b = -m$ gives a pair of integers not both zero, so $x \in B$.

Next prove $B \subseteq A$. If $x \in B$, then $ax + b = 0$ for integers a, b not both zero. If $a = 0$, this equation would force $b = 0$, a contradiction. Therefore $a \neq 0$, and

$$x = -\frac{b}{a} \in \mathbb{Q} = A.$$

Both inclusions hold, and hence $A = B$.

2 Logic and methods of proof

2.1 Quantifiers and implication

A *proposition* is a statement that is true or false. A formula with an unspecified variable may become a proposition only after we specify the variable or quantify it.

The symbol $\forall$ means "for every," and $\exists$ means "there exists." For example,

$$\forall x \in \mathbb{R} \quad x^2 \geq 0$$

is the true statement that every real square is nonnegative. The domain is essential. The usual real order relation $\geq$ is not defined on $\mathbb{C}$, so replacing $\mathbb{R}$ by $\mathbb{C}$ does not give a well-defined ordered statement. Moreover, $i^2 = -1$ shows directly that a complex square can be a negative real number.

The symbol $\Rightarrow$ means "implies." A statement $P \Rightarrow Q$ says that whenever P holds, Q must hold too.

Example 2.1 (One counterexample is enough)**.** Consider

$$\forall x \in \mathbb{R} \quad (x < 3 \Rightarrow x^2 < 3).$$

This is false. Take $x = -4$: the premise $x < 3$ is true, but the conclusion $x^2 < 3$ is false because $x^2 = 16$.

2.2 The order of quantifiers

The order of quantifiers can change a statement completely. In the following examples the inequality is $x + n > 3$.

Statement		Truth value
$\forall x \in \mathbb{R}\ \forall n \in \mathbb{Z}$	$x + n > 3$	False
$\forall n \in \mathbb{Z}\ \forall x \in \mathbb{R}$	$x + n > 3$	False
$\forall x \in \mathbb{R}\ \exists n \in \mathbb{Z}$	$x + n > 3$	True
$\exists n \in \mathbb{Z}\ \forall x \in \mathbb{R}$	$x + n > 3$	False
$\forall n \in \mathbb{Z}\ \exists x \in \mathbb{R}$	$x + n > 3$	True
$\exists x \in \mathbb{R}\ \forall n \in \mathbb{Z}$	$x + n > 3$	False

Here is why. The first two statements fail already at $x = 0$, $n = 0$. For any chosen x, we can choose an integer $n > 3 - x$, so the third statement is true. The fourth statement would require one fixed integer n to work for every x, but $x = 3 - n$ defeats that choice. For the fifth statement choose $x = 4 - n$. For the last statement, any proposed x can be defeated by choosing an integer $n \leq 3 - x$.

Two consecutive universal quantifiers can be exchanged, as can two consecutive existential quantifiers over fixed domains. In general, a universal and an existential quantifier cannot be exchanged. In $\forall x\, \exists n$, the choice of n may depend on x; in $\exists n\, \forall x$, the same n must work for every x.

Example 2.2 (A study room for each student, or one room for everyone?)**.** Suppose each of three students has an assigned study room: student 1 uses room A, student 2 uses room B, and student 3 uses room C. Then "for every student there exists an assigned room" is true. But "there exists a room assigned to every student" is false. The individual choices do not produce a common choice.

2.3 And, or, and not

The logical connective $\wedge$ means "and," and $\vee$ means inclusive "or." Their meanings are specified by truth tables. We use T for true and F for false.

P	Q	$P \wedge Q$	$P \vee Q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

Example 2.3 (A formula involving the integer part)**.** Let $\lfloor x \rfloor$ denote the greatest integer less than or equal to x. The formula

$$\forall x \in \mathbb{R}\ \exists n \in \mathbb{N} \quad (\lfloor x \rfloor = n \ \wedge\ n \leq x)$$

says that every real number has a *natural* integer part that is no larger than the number itself. It is false: for $x = -0.4$, we have $\lfloor x \rfloor = -1 \notin \mathbb{N}$.

The symbol $\neg$ means "not." Its truth table is

P	$\neg P$
T	F
F	T

The phrase "there does not exist a natural number" is incomplete unless we specify a property. The expression $\neg(\exists n \in \mathbb{N})$ alone does not state which condition n is supposed to satisfy.

Example 2.4 (Fermat's last theorem as a logical statement)**.** Fermat's last theorem says

$$\forall n \in \mathbb{N} \quad \left(n > 2 \Rightarrow \neg\left(\exists x, y, z \in \mathbb{N} \quad x^n + y^n = z^n\right)\right).$$

The abbreviated notation $\exists x, y, z \in \mathbb{N}$ means

$$\exists x \in \mathbb{N}\ \exists y \in \mathbb{N}\ \exists z \in \mathbb{N}.$$

In words, when the integer exponent is greater than 2, the equation has no solution in positive integers.

For comparison, when $n = 2$ there are positive integer solutions. For instance,

$$5^2 + 12^2 = 25 + 144 = 169 = 13^2.$$

Such triples are called *Pythagorean triples*.

Saying that there is no triple satisfying an equality is equivalent to saying that every triple fails to satisfy it:

$$\neg\left(\exists x, y, z \in \mathbb{N} \quad x^n + y^n = z^n\right) \equiv \forall x, y, z \in \mathbb{N} \quad x^n + y^n \neq z^n.$$

Thus the theorem can also be written

$$\forall n \in \mathbb{N} \quad \left(n > 2 \Rightarrow \left(\forall x, y, z \in \mathbb{N} \quad x^n + y^n \neq z^n\right)\right).$$

2.4 Logical equivalence and negation rules

We write $P \equiv Q$ when the formulas have the same truth value under every assignment of truth values to their basic statements. This is *logical equivalence*, which can be checked by comparing truth tables.

The main negation rules are

$$\begin{aligned} \neg(\forall x \in X\ P(x)) &\equiv \exists x \in X\ \neg P(x), \\ \neg(\exists x \in X\ P(x)) &\equiv \forall x \in X\ \neg P(x), \\ \neg(P \Rightarrow Q) &\equiv P \wedge \neg Q, \\ \neg(P \wedge Q) &\equiv (\neg P) \vee (\neg Q), \\ \neg(P \vee Q) &\equiv (\neg P) \wedge (\neg Q). \end{aligned}$$

The last two are *De Morgan's laws*. When a negation passes through a quantifier, $\forall$ and $\exists$ exchange roles, and the property inside must be negated too.

For implication, the truth table is

P	Q	$P \Rightarrow Q$	$\neg(P \Rightarrow Q)$
T	T	T	F
T	F	F	T
F	T	T	F
F	F	T	F

An implication fails exactly when its premise is true and its conclusion is false. This is why its negation is $P \wedge \neg Q$.

Example 2.5 (Negating a quantified formula)**.** The formula

$$\exists n \in \mathbb{Z}\ \forall x \in \mathbb{R} \quad x + n > 3$$

is false. Negating it gives

$$\forall n \in \mathbb{Z}\ \exists x \in \mathbb{R} \quad x + n \leq 3.$$

This is true: after n is given, choose $x = 3 - n$.

Similarly, the negation of Fermat's last theorem is

$$\exists n \in \mathbb{N} \quad \left(n > 2 \ \wedge\ \left(\exists x, y, z \in \mathbb{N} \quad x^n + y^n = z^n\right)\right).$$

It would assert the existence of an exponent greater than 2 and a positive integer solution for that exponent. This negation is false.

Finally, if

$$P: \quad \forall x \in \mathbb{R}\ \exists n \in \mathbb{N} \quad \lfloor x \rfloor = n,$$

then

$$\neg P: \quad \exists x \in \mathbb{R}\ \forall n \in \mathbb{N} \quad \lfloor x \rfloor \neq n.$$

For example, $x = -0.4$ is a witness to $\neg P$, because its integer part is -1.

Example 2.6 (Negation and set operations)**.** For an element x in a fixed universe U,

$$\begin{aligned} x \in (A \cup B)^c &\iff \neg(x \in A \text{ or } x \in B) \\ &\iff (x \notin A \text{ and } x \notin B) \\ &\iff x \in A^c \cap B^c. \end{aligned}$$

Thus $(A \cup B)^c = A^c \cap B^c$. The logical rule translates directly into a rule for sets.

2.5 Proof by contraposition

The implication $P \Rightarrow Q$ is logically equivalent to its *contrapositive*, $\neg Q \Rightarrow \neg P$:

$$(P \Rightarrow Q) \equiv (\neg Q \Rightarrow \neg P).$$

The full truth table verifies the equivalence:

P	Q	$\neg P$	$\neg Q$	$P \Rightarrow Q$	$\neg Q \Rightarrow \neg P$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Consequently, proving the contrapositive is a valid way to prove the original implication. It is often easier, although not always.

Example 2.7 (The converse is a different statement)**.** Every square is a rectangle. Equivalently, a shape that is not a rectangle cannot be a square. However, the converse "every rectangle is a square" is false: a rectangle with side lengths 2 and 5 is a counterexample.

Proposition 2.1 (A prime dividing a square)**.** If p is prime and $n \in \mathbb{N}$, then

$$p \mid n^2 \quad \Longrightarrow \quad p \mid n.$$

Here $p \mid n$ means that $n = pk$ for an integer k.

Proof. We prove the contrapositive. Suppose $p \nmid n$. If $n = 1$, then $p \nmid n^2 = 1$. If $n > 1$, use the elementary unique prime factorization theorem: write

$$n = q_1 q_2 \dots q_r$$

with each q_j prime, allowing repetitions. No factor q_j equals p, since $p \nmid n$. Then

$$n^2 = q_1 q_1 q_2 q_2 \dots q_r q_r$$

has no prime factor equal to p, so $p \nmid n^2$. This proves the contrapositive and hence the claim. □

For example, if $5 \mid n^2$, then $5 \mid n$. In particular, if a square is even, then its integer square root is even. The same prime-divisibility statement applies to integers, since replacing an integer by its absolute value changes neither its square nor its divisibility by a positive prime.

2.6 Proof by contradiction

To prove a proposition S by contradiction, assume $\neg S$ and derive an impossibility. Then the assumption $\neg S$ must be false. To prove an implication $P \Rightarrow Q$, its negation is $P \wedge \neg Q$, so a contradiction proof begins by assuming both P and the failure of Q.

A proof by contraposition assumes $\neg Q$ and proves $\neg P$. A proof by contradiction assumes the negation of the entire desired statement and reaches a false conclusion. The two methods are related, but their written structure need not be identical.

Example 2.8 (The irrationality of $\sqrt{3}$)**.** We prove that no rational number r satisfies $r^2 = 3$. Suppose, for a contradiction, that such an r exists. Write it in canonical form,

$$r = \frac{m}{n}, \qquad m \in \mathbb{Z}, \quad n \in \mathbb{N}, \quad \gcd(m, n) = 1.$$

Then

$$\frac{m^2}{n^2} = 3 \quad \Longrightarrow \quad m^2 = 3n^2.$$

Thus $3 \mid m^2$, and the prime-divisibility proposition gives $3 \mid m$. Write $m = 3k$ with $k \in \mathbb{Z}$. Substituting,

$$3n^2 = m^2 = (3k)^2 = 9k^2,$$

so $n^2 = 3k^2$. Again the proposition gives $3 \mid n$. Therefore both m and n are divisible by 3, contradicting $\gcd(m, n) = 1$.

There is no such rational r. In particular, the positive real number $\sqrt{3}$ is irrational.

3 Induction and the binomial theorem

3.1 The induction principle

The principle of mathematical induction says that if a subset $A \subseteq \mathbb{N}$ satisfies

$$1 \in A, \qquad n \in A \Rightarrow n+1 \in A \quad (n \in \mathbb{N}),$$

then $A = \mathbb{N}$.

For a sequence of propositions $P(1), P(2), \ldots$, it gives the following proof method:

1. *Base case*: prove $P(1)$.
2. *Inductive step*: for an arbitrary $n \in \mathbb{N}$, assume $P(n)$ and use that assumption to prove $P(n+1)$.

The temporary assumption $P(n)$ in the second step is the *induction hypothesis*. We do not assume the result for every n at once. We prove the implication from any one case to the next.

Example 3.1 (The sum of the first n natural numbers)**.** We prove

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} \qquad (n \in \mathbb{N}).$$

For $n = 1$, the right-hand side is $1(1+1)/2 = 1$, so the base case holds.

Assume the formula holds for n. Then

$$\begin{aligned} 1 + 2 + \cdots + n + (n+1) &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{n(n+1) + 2(n+1)}{2} \\ &= \frac{(n+1)(n+2)}{2}. \end{aligned}$$

This is the desired formula for $n+1$. Induction proves the result for every positive integer n.

Example 3.2 (Arithmetic sequences)**.** A sequence $a_1, a_2, \ldots$ is *arithmetic* if there is a fixed $d \in \mathbb{R}$ such that $a_{n+1} = a_n + d$ for every $n \in \mathbb{N}$. We claim

$$a_n = a_1 + (n-1)d.$$

For $n = 1$, the right side is $a_1 + (1-1)d = a_1$. If the formula holds for n, then

$$a_{n+1} = a_n + d = a_1 + (n-1)d + d = a_1 + nd,$$

which is the formula for the next term. For example, if $a_1 = 7$ and $d = 3$, then $a_n = 3n + 4$ and $a_8 = 28$.

Example 3.3 (Geometric sequences)**.** A sequence is *geometric* if $a_{n+1} = qa_n$ for a fixed real number q. Its terms satisfy

$$a_n = a_1 q^{n-1}.$$

For the base case, interpret $q^0 = 1$, including the first term when $q = 0$; equivalently one can state the first term separately and apply the formula for $n \geq 2$. Thus the formula at $n = 1$ gives a_1. Assuming the formula for n,

$$a_{n+1} = qa_n = q(a_1 q^{n-1}) = a_1 q^n.$$

This establishes the inductive step. For $a_1 = 3$, $q = 2$, the first four terms are $3, 6, 12, 24$.

Example 3.4 (Odd numbers build squares)**.** We prove that the first n positive odd numbers sum to n^2:

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$

For $n = 1$, both sides equal 1. If the formula holds for n, adding the next odd number gives

$$n^2 + (2n + 1) = n^2 + 2n + 1 = (n + 1)^2.$$

This proves the next case. For example, $1 + 3 + 5 + 7 + 9 = 25$. The identity also explains why the difference between consecutive square numbers is always odd.

3.2 Checking the starting point

A proof by induction may begin at a number n_0 other than 1. In that case, prove $P(n_0)$ and the implication $P(n) \Rightarrow P(n + 1)$ for every $n \geq n_0$. The conclusion concerns only $n \geq n_0$.

Example 3.5 (The false proof that all horses have the same color)**.** Consider the claim that every group of n horses consists of horses of the same color. The base case $n = 1$ is true. A tempting "inductive step" takes $n + 1$ horses and compares the first n horses with the last n horses. By the induction hypothesis, each group has a single color. If the groups share a horse, their colors must match.

The gap is at $n = 1$. The first one-horse group and the last one-horse group in a group of two horses do not overlap. No shared horse connects their colors. Thus the argument does not prove the step from 1 to 2, and the induction fails.

Example 3.6 (An inequality starting at $n = 3$)**.** For all integers $n \geq 3$,

$$2n + 1 < 2^n.$$

The base case is $2(3) + 1 = 7 < 8 = 2^3$. Suppose $n \geq 3$ and $2n + 1 < 2^n$. Then

$$\begin{aligned} 2(n + 1) + 1 &= (2n + 1) + 2 \\ &< 2^n + 2 \\ &< 2^n + 2^n = 2^{n+1}, \end{aligned}$$

where the second strict inequality uses $2 < 2^n$. The result follows by induction.

Example 3.7 (An inequality starting at $n = 5$)**.** For all integers $n \geq 5$,

$$n^2 < 2^n.$$

For $n = 5$, this is $25 < 32$. Assume it holds at some $n \geq 5$. Using the preceding inequality,

$$\begin{aligned} (n + 1)^2 &= n^2 + 2n + 1 \\ &< 2^n + 2n + 1 \\ &< 2^n + 2^n = 2^{n+1}. \end{aligned}$$

This completes the step. Although $1^2 < 2^1$ is also true, it cannot be used with this argument to conclude the result for every $n \in \mathbb{N}$: the required step does not hold at all the smaller values. In fact, $3^2 < 2^3$ is false.

3.3 Summation notation

For terms $a_1, \ldots, a_n$, the symbol

$$\sum_{k=1}^{n} a_k = a_1 + a_2 + \cdots + a_n$$

is a concise way to write a sum. The index k tells us which term is being added; it is a temporary index and can be renamed consistently.

Example 3.8.

$$\sum_{k=1}^{13} 1 = 13,$$
$$\sum_{k=1}^{8} k = 1+2+3+4+5+6+7+8 = 36,$$
$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}.$$

Also,

$$\sum_{k=1}^{n} (-1)^k = \begin{cases} 0, & n \text{ is even,} \\ -1, & n \text{ is odd.} \end{cases}$$

Indeed, consecutive pairs $(-1)+1$ cancel; an odd number of terms leaves one final -1.

Sums can be nested. In

$$\sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij},$$

first add the terms with $j = 1, \ldots, m$ for each fixed i, and then add the resulting n sums. For instance, if $a_{11} = 2$, $a_{12} = 3$, $a_{21} = 4$, and $a_{22} = 1$, then

$$\sum_{i=1}^{2} \sum_{j=1}^{2} a_{ij} = (2+3) + (4+1) = 10.$$

3.4 Newton's binomial theorem

Expanding two factors gives

$$(a+b)^2 = (a+b)(a+b) = a^2 + ab + ba + b^2 = a^2 + 2ab + b^2.$$

Similarly,

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3.$$

Each term has total exponent equal to the number of factors. The question is how many times each product appears.

In $(a+b)^n$, we choose either a or b from each of the n factors. To obtain $a^k b^{n-k}$, choose exactly k of the factors to contribute a. There are

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

such choices. Here $n! = 1 \times 2 \times \cdots \times n$, and $0! = 1$. The number $\binom{n}{k}$ is read "n choose k" and counts the k-element subsets of an n-element set. One explanation of the formula is that ordered choices of k distinct objects give $n!/(n-k)!$ possibilities, and each unordered selection is counted $k!$ times.

Theorem 3.1 (The binomial theorem)**.** For $a, b \in \mathbb{R}$ and $n \in \mathbb{N}$,

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}.$$

An exponent zero denotes the empty product, whose value in this polynomial identity is 1.

The counting argument above proves the formula by expansion. We can also prove it by induction.

Inductive proof. First note the identity

$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k} \qquad (1 \le k \le n).$$

Indeed, putting the factorial expressions over a common denominator gives

$$\begin{aligned} \frac{n!}{(k-1)!(n-k+1)!} + \frac{n!}{k!(n-k)!} &= \frac{n!k + n!(n-k+1)}{k!(n+1-k)!} \\ &= \frac{n!(n+1)}{k!(n+1-k)!} \\ &= \frac{(n+1)!}{k!(n+1-k)!}. \end{aligned}$$

This is Pascal's identity.

For $n = 1$, the proposed sum is $b + a = a + b$. Suppose the binomial formula holds for n. Multiplication by $a + b$ gives

$$\begin{aligned} (a+b)^{n+1} &= a \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k} + b \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k} \\ &= \sum_{k=0}^{n} \binom{n}{k} a^{k+1} b^{n-k} + \sum_{k=0}^{n} \binom{n}{k} a^k b^{n+1-k} \\ &= a^{n+1} + b^{n+1} + \sum_{k=1}^{n} \left(\binom{n}{k-1} + \binom{n}{k} \right) a^k b^{n+1-k} \\ &= a^{n+1} + b^{n+1} + \sum_{k=1}^{n} \binom{n+1}{k} a^k b^{n+1-k} \\ &= \sum_{k=0}^{n+1} \binom{n+1}{k} a^k b^{n+1-k}. \end{aligned}$$

The middle terms combine after shifting the index in the first sum; the two endpoint terms each have coefficient 1. This completes the induction. □

The symmetry

$$\binom{n}{k} = \binom{n}{n-k}$$

is immediate from the factorial formula: choosing which k positions supply a is equivalent to choosing which $n - k$ positions supply b.

Example 3.9 (The fifth power)**.** The coefficients are

$$\begin{aligned} &\binom{5}{0} = 1, && \binom{5}{1} = 5, && \binom{5}{2} = \frac{5!}{2!3!} = 10, \\ &\binom{5}{3} = 10, && \binom{5}{4} = 5, && \binom{5}{5} = 1. \end{aligned}$$

Hence

$$(a+b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5.$$

Substituting $a = b = 1$ gives $32 = 1 + 5 + 10 + 10 + 5 + 1$.

Example 3.10 (Adding an entire row of binomial coefficients)**.** Set $a = b = 1$ in the general formula:

$$\sum_{k=0}^{n} \binom{n}{k} = \sum_{k=0}^{n} \binom{n}{k} 1^k 1^{n-k} = (1+1)^n = 2^n.$$

This also counts all subsets of an n-element set, grouped by their size. Each element can independently be included or excluded, giving two choices per element.

4 Fields

4.1 Why abstract the rules of arithmetic?

The rational and real numbers permit division by every nonzero number. The natural numbers and the integers do not. A *field* is a set with two operations that satisfy the familiar rules making such arithmetic possible.

The symbols for the operations are part of the structure. They need not mean ordinary addition and multiplication of real numbers. In this section, explicit multiplication in the field is written with the cross symbol $\times$; juxtaposition means the same multiplication when it is unambiguous.

Definition 4.1 (Field)**.** A *field* is a set F together with two binary operations, denoted $+$ and $\times$, satisfying the following requirements.

1. **Binary operations.** For every $a, b \in F$, each operation assigns a uniquely determined result in F: both $a + b \in F$ and $a \times b \in F$.
2. **Associativity.** For every $a, b, c \in F$,
$$a + (b + c) = (a + b) + c, \qquad a \times (b \times c) = (a \times b) \times c.$$
3. **Commutativity.** For every $a, b \in F$,
$$a + b = b + a, \qquad a \times b = b \times a.$$
4. **Distinct identities.** There are distinct elements $0, 1 \in F$ such that for every $a \in F$,
$$a + 0 = a, \qquad a \times 1 = a.$$
5. **Additive inverses.** For every $a \in F$, there is a $b \in F$ with $a + b = 0$.
6. **Multiplicative inverses.** For every $a \in F$ with $a \neq 0$, there is a $b \in F$ with $a \times b = 1$.
7. **Distributivity.** For every $a, b, c \in F$,
$$a \times (b + c) = a \times b + a \times c.$$

We may denote the entire structure by $(F, +, \times)$.

Calling the operations "binary operations on F" already includes that their results are well defined and belong to F. The word *closed* is useful when restricting existing operations to a subset: the subset is closed under the operation if its result remains in the subset. Merely checking closure does not check all the field axioms.

The elements called 0 and 1 are the identities of the operations. In an abstract field these names express their algebraic roles; we do not assume that the field is a subset of $\mathbb{R}$.

4.2 Basic examples and nonexamples

Example 4.1. 1. With ordinary addition and multiplication, $\mathbb{N}$ is not a field. It has no additive identity within $\mathbb{N}$ under our convention $0 \notin \mathbb{N}$.

2. With the usual operations, $\mathbb{Z}$ is not a field. The multiplicative identity must be the integer 1, since an identity e must satisfy $1 \times e = 1$. But there is no integer k satisfying $3 \times k = 1$, so 3 has no inverse in $\mathbb{Z}$.
3. The rational numbers $\mathbb{Q}$, with ordinary addition and multiplication, form a field. The identities are 0 and 1. The additive inverse of m/n is $-m/n$, and if $m \neq 0$, its multiplicative inverse is n/m. All the associativity, commutativity, and distributivity rules are inherited from rational arithmetic.
4. The real numbers $\mathbb{R}$, with ordinary addition and multiplication, form a field as well.

Even when the underlying objects are familiar numbers, one may define different operations on them and ask whether the field rules still hold. The answer depends on both the set and the operations.

4.3 Constructing a field with two elements

Suppose $F = \{a, b\}$ has exactly two elements. A field must have two distinct identities, so one element must be the additive identity and the other the multiplicative identity. Call them $a = 0$ and $b = 1$.

The identity requirements and the inverse requirement force the following tables:

$+$	a	b
a	a	b
b	b	a

$\times$	a	b
a	a	a
b	a	b

For example, $b + b$ cannot be b: otherwise neither element would be an additive inverse for b. Hence $b + b = a$. Multiplication by b leaves both elements unchanged. Also, $a \times a = a$ follows from distributivity. Indeed,

$$a \times (a + b) = (a \times a) + (a \times b)$$

has left side $a \times b = a$ and right side $(a \times a) + a = a \times a$. Thus $a \times a = a$.

These operations really do satisfy the field axioms. One quick check is to label a by 0 and b by 1: addition is ordinary addition followed by taking the even or odd remainder, and multiplication is the corresponding product. Associativity and distributivity follow by expanding the integer expressions and noting that equal integers have the same parity. Both identities are present, each element is its own additive inverse, and the only nonzero element b is its own multiplicative inverse.

In particular,

$$1 + 1 = 0$$

in this field. This is consistent because addition here is given by its table, rather than ordinary real addition.

Practice 4.1. Try to build addition and multiplication tables for fields with 3, 4, 5, and 6 elements. The table entries and the operations are yours to choose. Check every field axiom; do not assume that ordinary remainders will work for every proposed size. The search may take some time, and not every requested size need succeed.

4.4 Consequences of the field axioms

Theorem 4.2 (Basic properties of a field)**.** Let F be a field.

1. The additive identity 0 is unique, and the multiplicative identity 1 is unique.
2. Every element a has a unique additive inverse, denoted $-a$. Every nonzero element a has a unique multiplicative inverse, denoted a^{-1}.
3. For every $a \in F$, $-(-a) = a$.
4. For every nonzero $a \in F$, $(a^{-1})^{-1} = a$.
5. For every $a \in F$, $a \times 0 = 0$.
6. If $a \times b = 0$, then $a = 0$ or $b = 0$.

Proof. **1. Uniqueness of the identities.** If u and v are both additive identities, then

$$u = u + v = v.$$

The first equality uses that v is an identity, and the second uses that u is an identity, together with commutativity. If e and f are both multiplicative identities, then similarly

$$e = e \times f = f.$$

2. Uniqueness of inverses. If $a + b = 0$ and $a + c = 0$, then

$$b = b + 0 = b + (a + c) = (b + a) + c = 0 + c = c.$$

If $a \neq 0$, $a \times b = 1$, and $a \times c = 1$, then

$$\begin{aligned} b &= b \times 1 = b \times (a \times c) \\ &= (b \times a) \times c = 1 \times c = c. \end{aligned}$$

Thus both inverse notations are unambiguous.

3. The negative of a negative. Since $a + (-a) = 0$, commutativity shows that a is an additive inverse of $-a$. Uniqueness gives $-(-a) = a$.

5. Multiplication by zero. We prove this before part 4 so that we can also justify that a multiplicative inverse is nonzero. Distributivity gives

$$a \times 0 = a \times (0 + 0) = (a \times 0) + (a \times 0).$$

Add the additive inverse of $a \times 0$ to both sides. The left side becomes 0 and the right side becomes $a \times 0$, so $a \times 0 = 0$.

4. The inverse of an inverse. If $a \neq 0$, then $a \times a^{-1} = 1$. Its inverse a^{-1} cannot be zero, since otherwise the left side would be $a \times 0 = 0$, contrary to $0 \neq 1$. Commutativity gives $a^{-1} \times a = 1$. Therefore a is the unique multiplicative inverse of a^{-1}, so $(a^{-1})^{-1} = a$.

6. No nonzero factors of zero. Suppose $a \times b = 0$. If $a = 0$, the conclusion already holds. Otherwise multiply by a^{-1}:

$$\begin{aligned} b &= 1 \times b = (a^{-1} \times a) \times b \\ &= a^{-1} \times (a \times b) = a^{-1} \times 0 = 0. \end{aligned}$$

Thus at least one factor is zero. □

Two nonzero elements whose product is zero are called *zero divisors*. The last property says that a field has no zero divisors. It also explains why a product equation can be split into cases.

Example 4.2 (A product equation in a field)**.** Over $\mathbb{R}$, solve $(x-2)(x+5)=0$. The no-zero-divisors property gives

$$x-2=0 \quad \text{or} \quad x+5=0,$$

so $x=2$ or $x=-5$. Checking the candidates in the original expression confirms that both are solutions. The reasoning uses a field property, not just the appearance of two factors.